US 20250272580A1

(19) United States

(12) Patent Application Publication (10) Pub. No.: US 2025/0272580 A1

Shai et al. (43) Pub. Date: Aug. 28, 2025

(54) SYSTEM AND METHOD FOR PARALLEL PROCESSING OF A DECISION TREE

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(21) Appl. No.: **18/590,064**

(22) Filed: **Feb. 28, 2024**

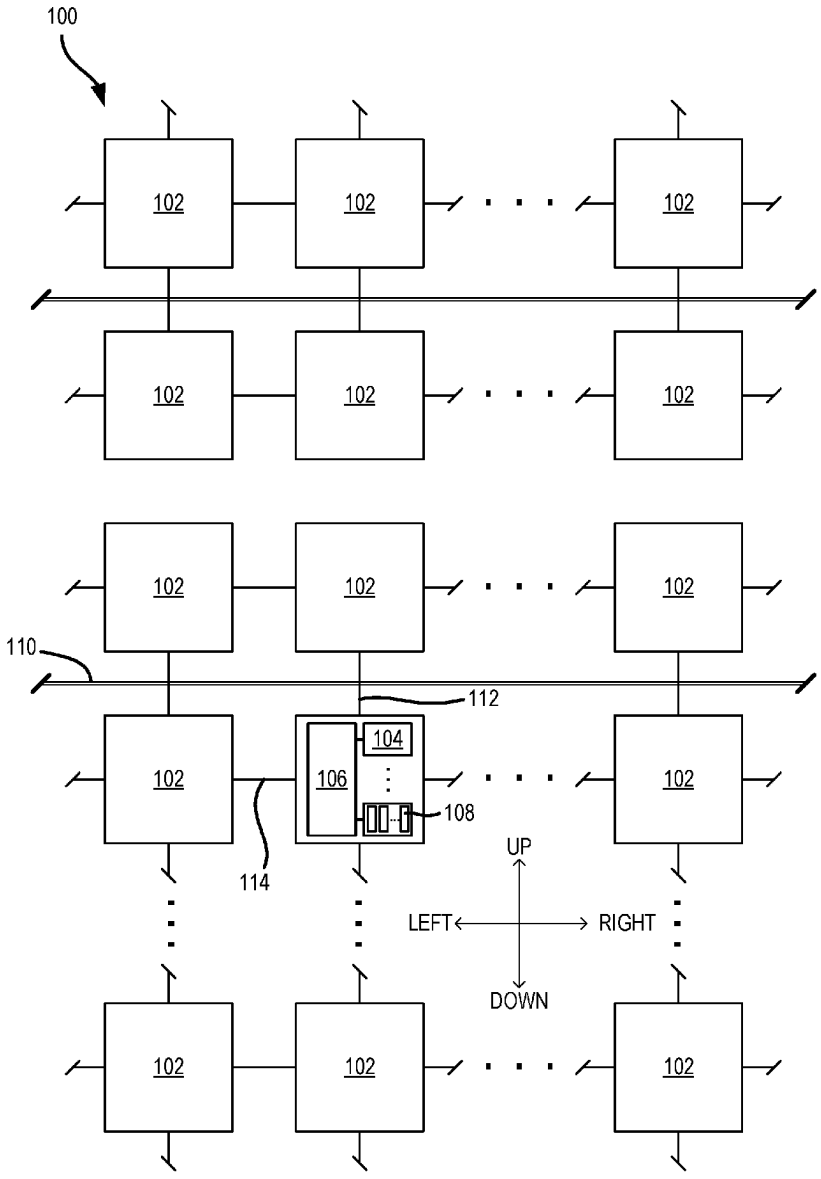
Publication Classification

(51) **Int. Cl.**
G06N 5/02 (2023.01)

(52) **U.S. Cl.**
CPC *G06N 5/02* (2013.01)

(57) **ABSTRACT**

An example computing device includes: a bank of processing elements; and a controller configured to: obtain an input vector having a plurality of input attributes, the input vector to be processed by a decision tree to identify a determined outcome; control the bank of processing elements to process the input vector to obtain a result vector, wherein each input attribute is processed by one of the processing elements in the bank to obtain a result, and wherein the result vector comprises a combination of the results; control the bank of processing elements to accumulate the result vector with an outcome vector for each potential outcome of the decision tree to obtain a respective outcome metric for each potential outcome; and select one potential outcome as the determined outcome of the decision tree for the input vector based on the respective outcome metrics for each potential outcome.



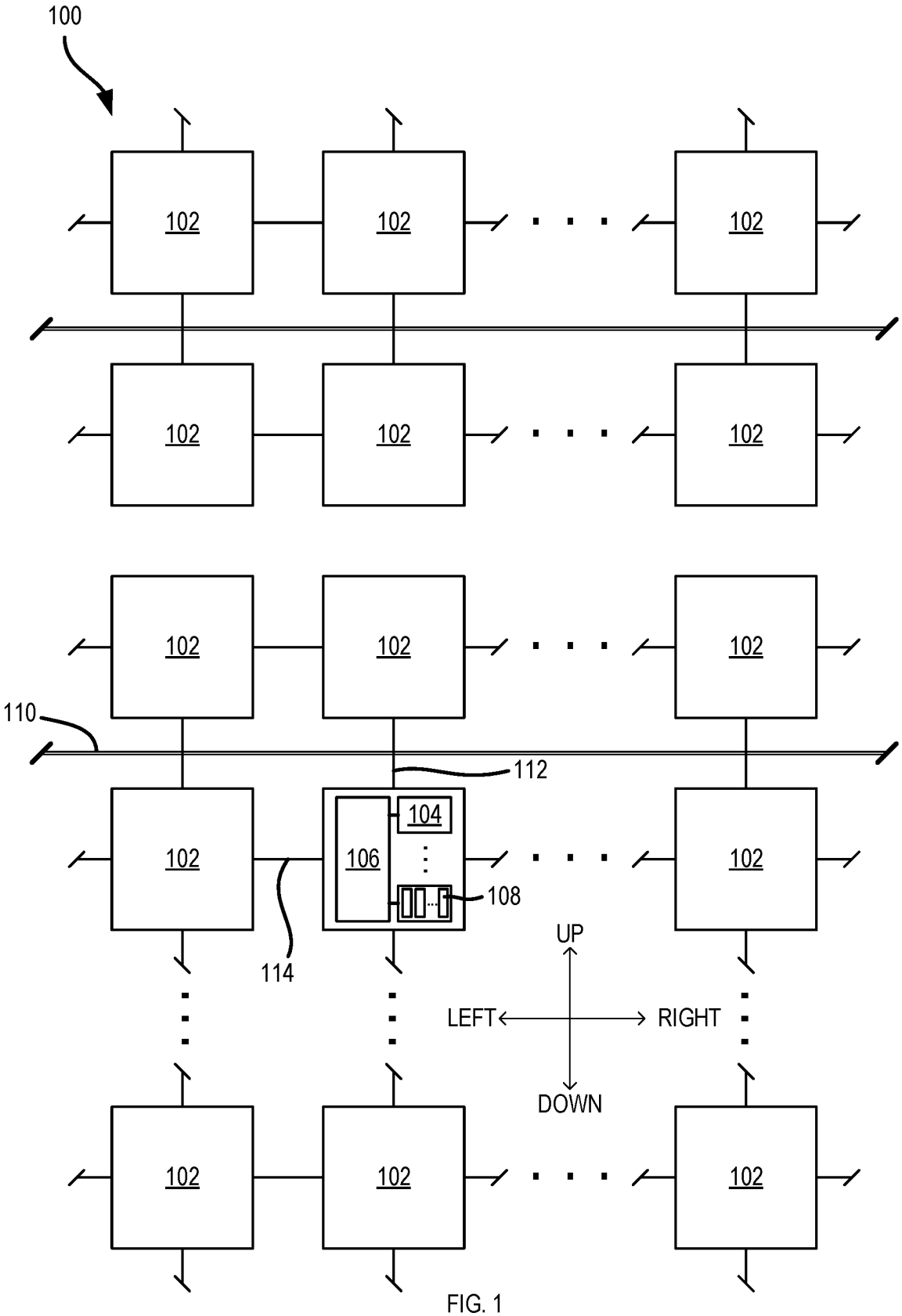


FIG. 1

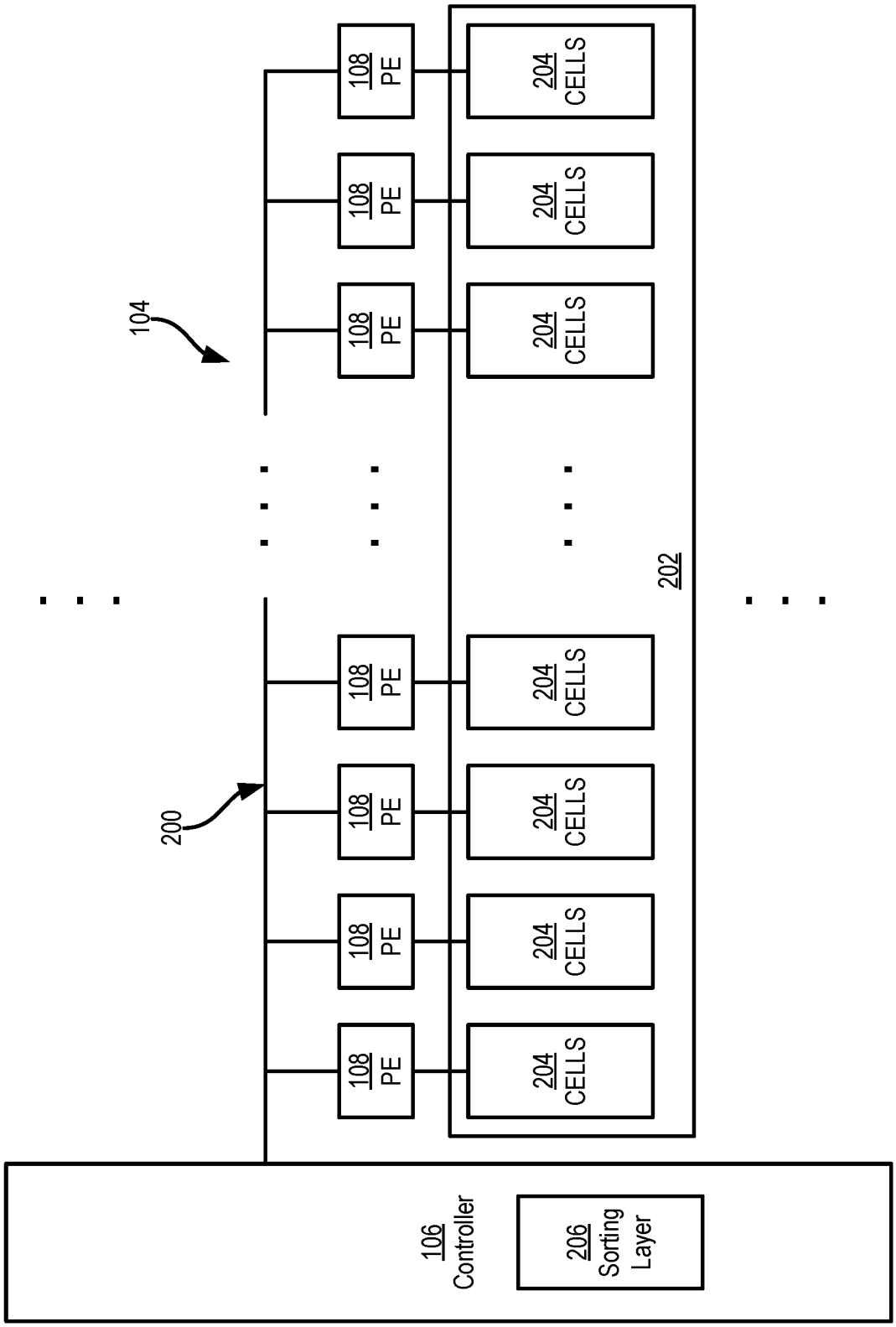

104
200
108 PE
204 CELLS
202
106 Controller
206 Sorting Layer

FIG. 2

$$\begin{bmatrix} d_0 \\ d_1 \\ d_2 \\ d_3 \end{bmatrix} = \begin{bmatrix} c_{00} & c_{01} & c_{02} & c_{03} \\ c_{10} & c_{11} & c_{12} & c_{13} \\ c_{20} & c_{21} & c_{22} & c_{23} \\ c_{30} & c_{31} & c_{32} & c_{33} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

FIG. 3A

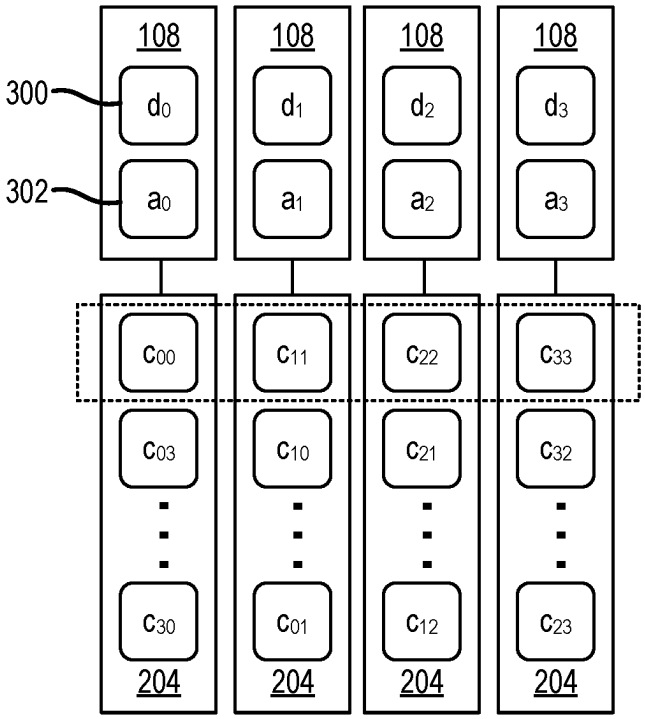


FIG. 3B

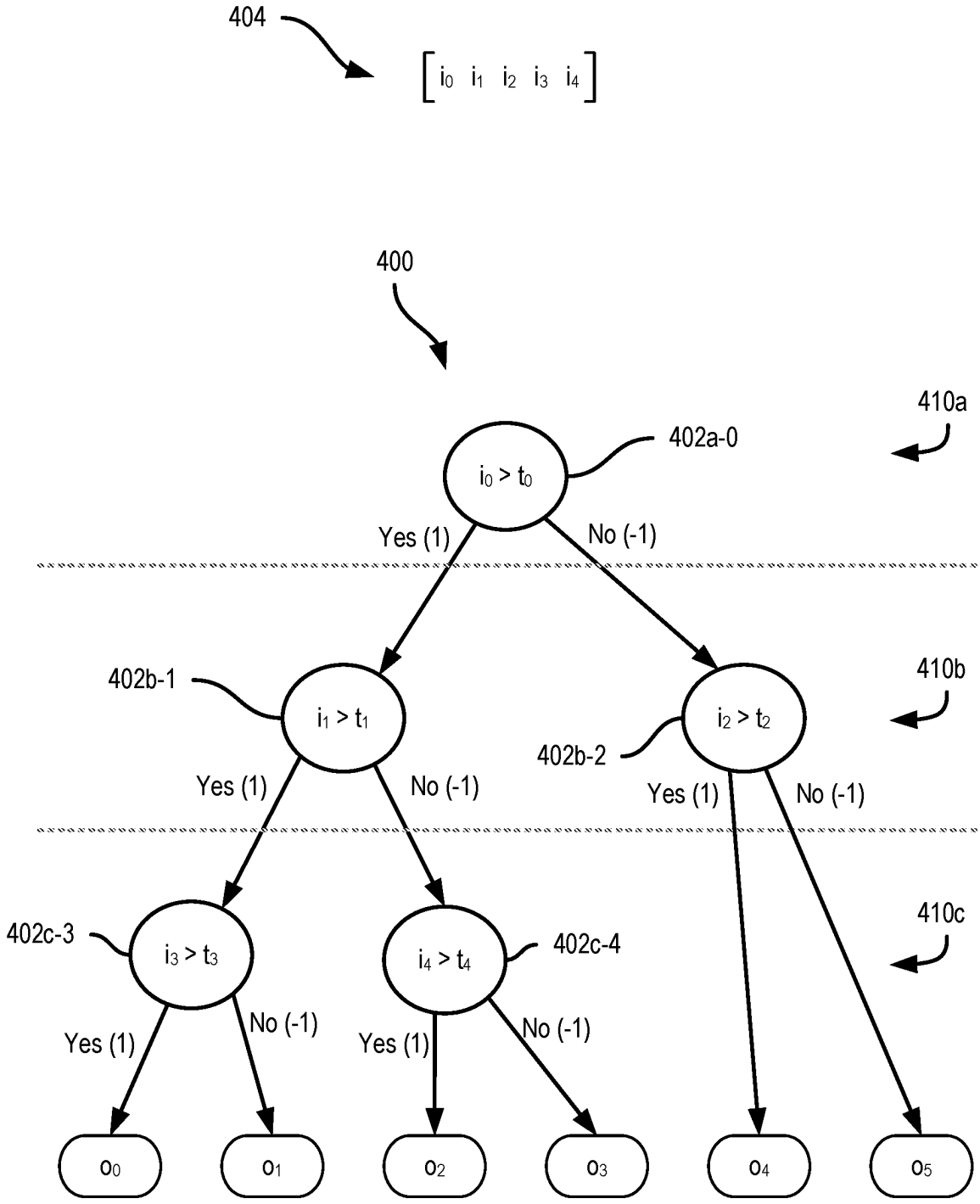

404
$[i_0\ i_1\ i_2\ i_3\ i_4]$
400
410a
402a-0
$i_0 > t_0$
Yes (1)
No (-1)
402b-1
$i_1 > t_1$
410b
$i_2 > t_2$
402b-2
Yes (1)
No (-1)
Yes (1)
No (-1)
402c-3
$i_3 > t_3$
$i_4 > t_4$
402c-4
410c
Yes (1)
No (-1)
Yes (1)
No (-1)
o_0
o_1
o_2
o_3
o_4
o_5

FIG. 4

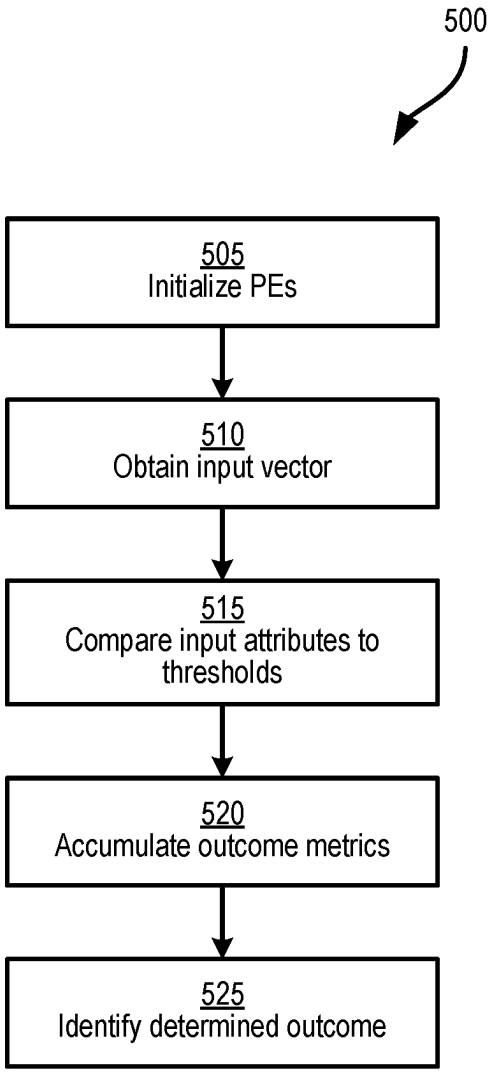

500
505
Initialize PEs
510
Obtain input vector
515
Compare input attributes to thresholds
520
Accumulate outcome metrics
525
Identify determined outcome

FIG. 5

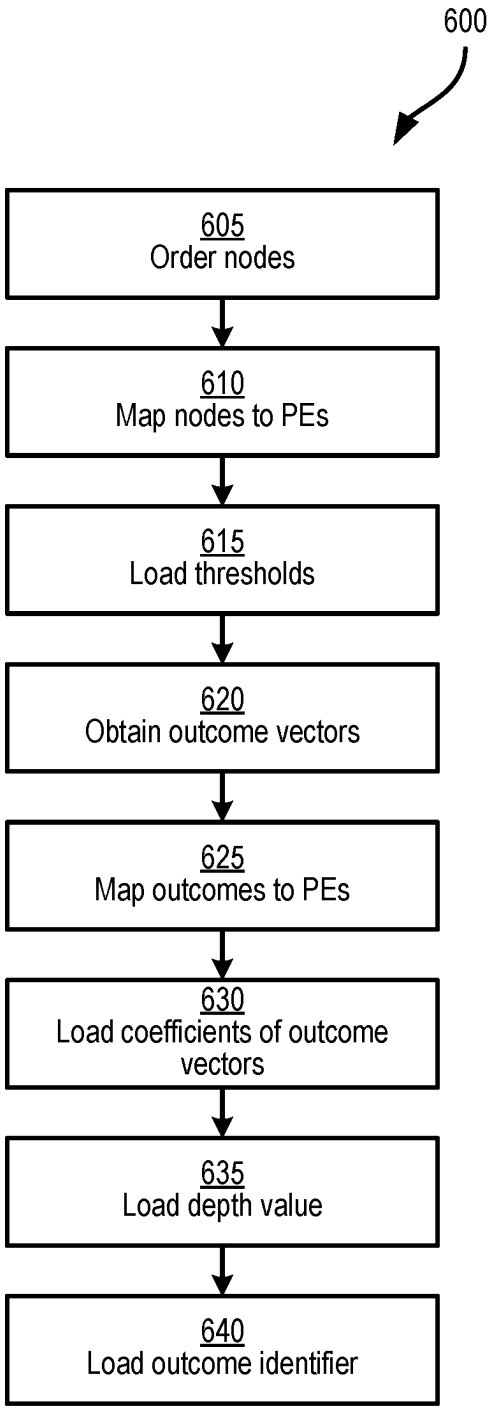

600
605
Order nodes
610
Map nodes to PEs
615
Load thresholds
620
Obtain outcome vectors
625
Map outcomes to PEs
630
Load coefficients of outcome vectors
635
Load depth value
640
Load outcome identifier

FIG. 6

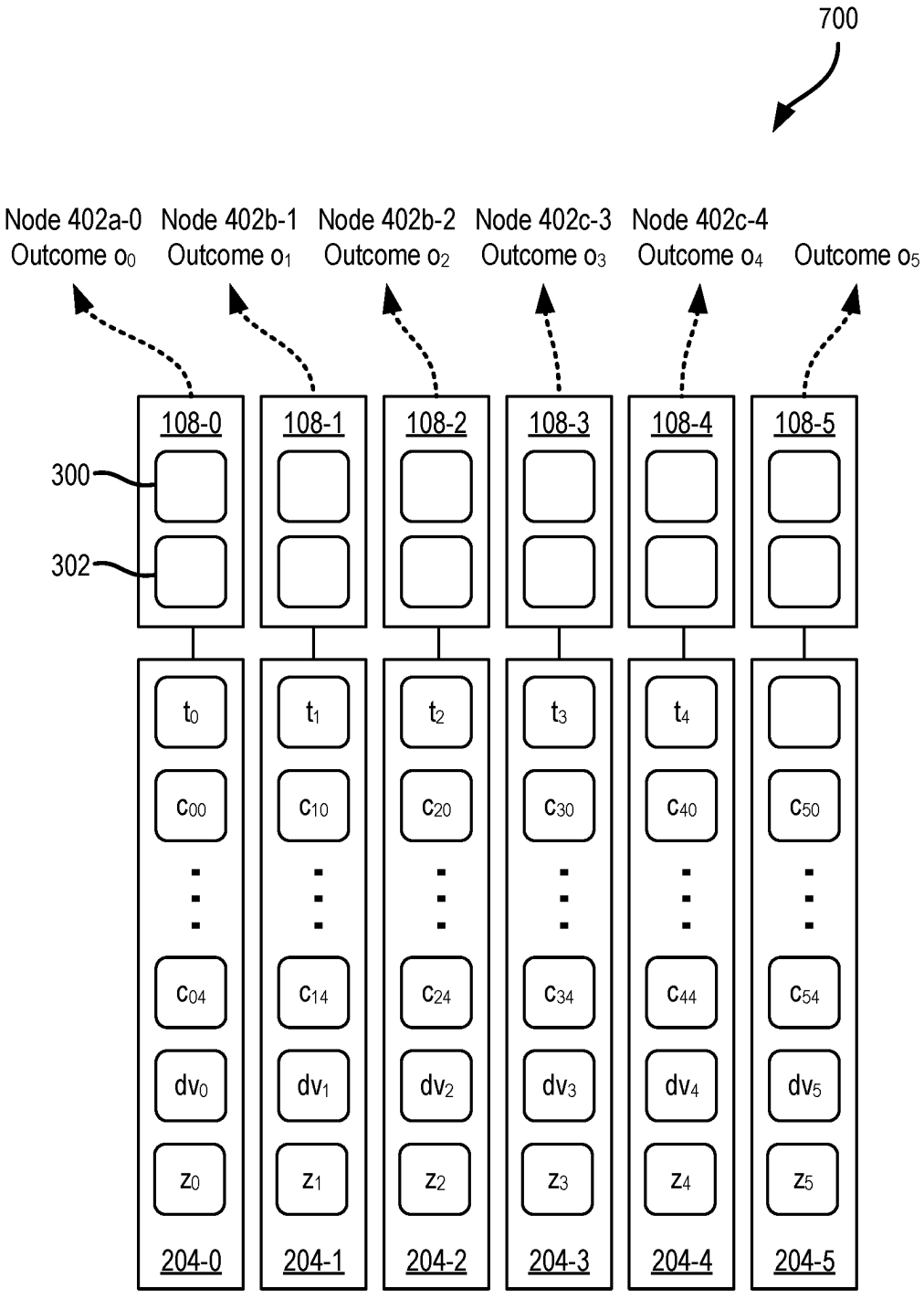

700
Node 402a-0
Outcome o_0
Node 402b-1
Outcome o_1
Node 402b-2
Outcome o_2
Node 402c-3
Outcome o_3
Node 402c-4
Outcome o_4
Outcome o_5
108-0
108-1
108-2
108-3
108-4
108-5
300
302
t_0
t_1
t_2
t_3
t_4
c_{00}
c_{10}
c_{20}
c_{30}
c_{40}
c_{50}
c_{04}
c_{14}
c_{24}
c_{34}
c_{44}
c_{54}
dv_0
dv_1
dv_2
dv_3
dv_4
dv_5
z_0
z_1
z_2
z_3
z_4
z_5
204-0
204-1
204-2
204-3
204-4
204-5

FIG. 7

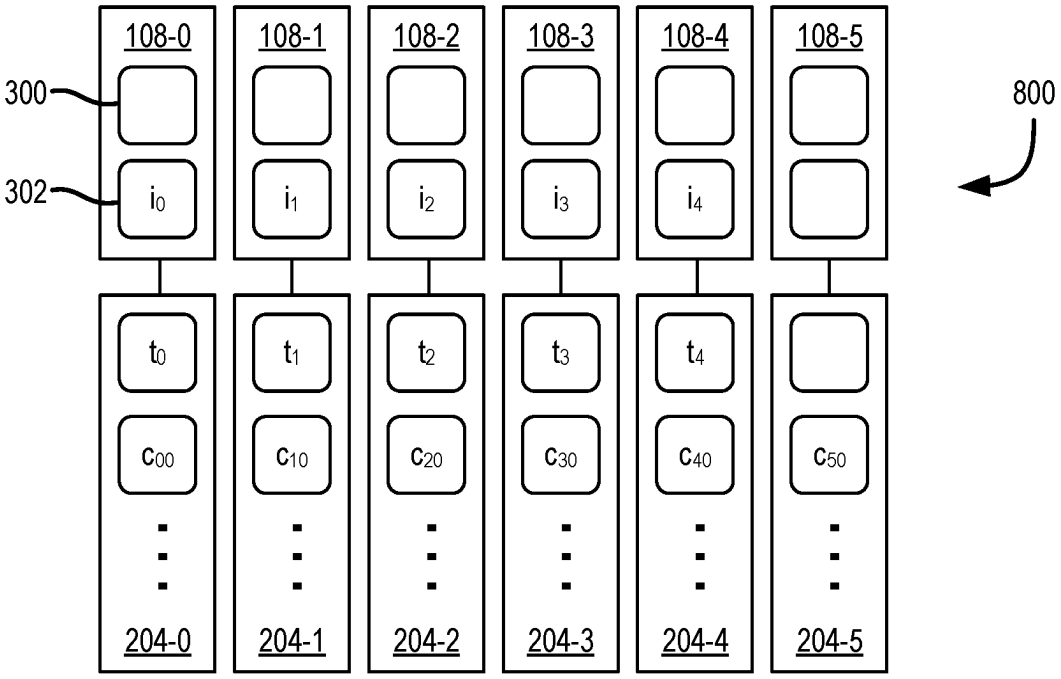

108-0
108-1
108-2
108-3
108-4
108-5
300
302
800
i_0
i_1
i_2
i_3
i_4
t_0
t_1
t_2
t_3
t_4
c_{00}
c_{10}
c_{20}
c_{30}
c_{40}
c_{50}
204-0
204-1
204-2
204-3
204-4
204-5

FIG. 8A

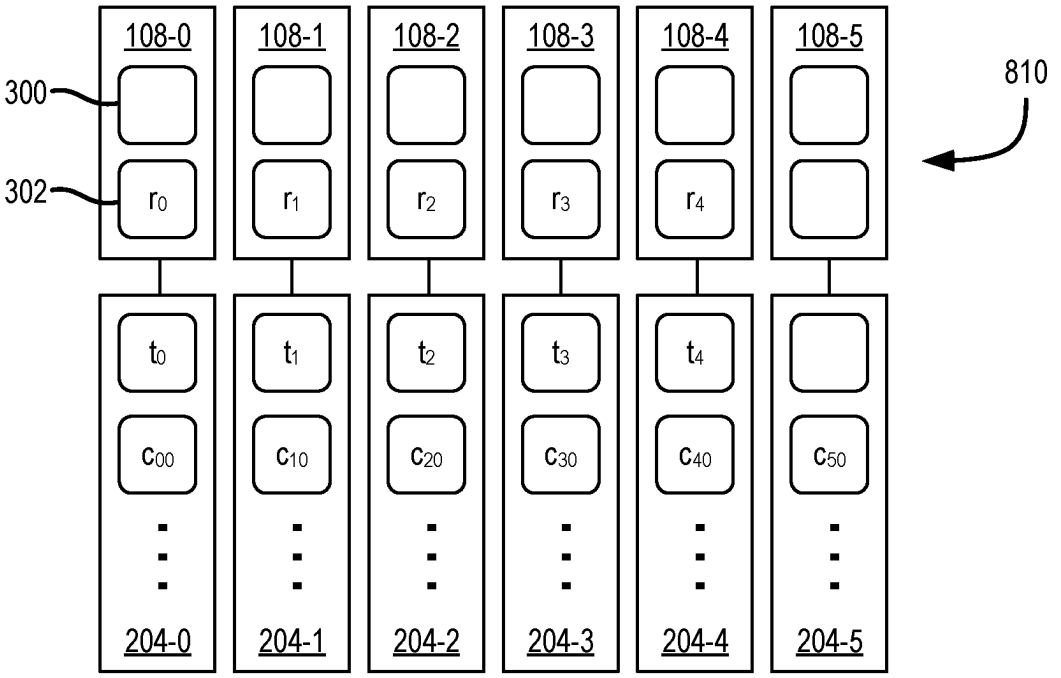

108-0
108-1
108-2
108-3
108-4
108-5
300
302
810
r_0
r_1
r_2
r_3
r_4
t_0
t_1
t_2
t_3
t_4
c_{00}
c_{10}
c_{20}
c_{30}
c_{40}
c_{50}
204-0
204-1
204-2
204-3
204-4
204-5

FIG. 8B

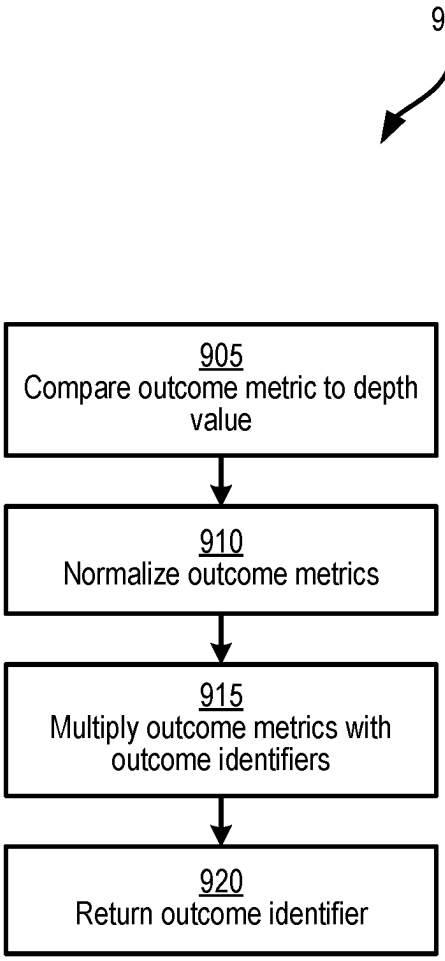

900
905
Compare outcome metric to depth value
910
Normalize outcome metrics
915
Multiply outcome metrics with outcome identifiers
920
Return outcome identifier

FIG. 9

SYSTEM AND METHOD FOR PARALLEL PROCESSING OF A DECISION TREE

FIELD

[0001] The specification relates generally to processing decision trees to determine an outcome of the decision tree, and more particularly to a system and method for parallel processing of a decision tree.

BACKGROUND

[0002] Decision trees may be used in many industries to select different outcomes from a plurality of potential outcomes based on a series of individual determinations based on a respective series of attributes at different nodes of the decision tree. The nodes of the decision tree are processed hierarchically, with given nodes being processed based on the results from an earlier level node. The sequential and hierarchical nature of decision trees results in slow and inefficient processing.

SUMMARY

[0003] According to an aspect of the present specification an example computing device includes: a bank of processing elements; a controller interconnected with the bank of processing elements, the controller configured to: obtain an input vector having a plurality of input attributes, the input vector to be processed by a decision tree to identify a determined outcome for the input vector; control the bank of processing elements to process the input vector to obtain a result vector, wherein each input attribute is processed by one of the processing elements in the bank to obtain a result, and wherein the result vector comprises a combination of the results; control the bank of processing elements to accumulate the result vector with an outcome vector for each potential outcome of the decision tree to obtain a respective outcome metric for each potential outcome; and select one potential outcome as the determined outcome of the decision tree for the input vector based on the respective outcome metrics for each potential outcome.

[0004] According to another aspect of the present specification, an example method includes: obtaining an input vector having a plurality of input attributes, the input vector to be processed by a decision tree to identify a determined outcome for the input vector; controlling a bank of processing elements to process the input vector to obtain a result vector, wherein each input attribute is processed by one of the processing elements in the bank to obtain a result, and wherein the result vector comprises a combination of the results; controlling the bank of processing elements to accumulate the result vector with an outcome vector for each potential outcome of the decision tree to obtain a respective outcome metric for each potential outcome; and selecting one potential outcome as the determined outcome of the decision tree for the input vector based on the respective outcome metrics for each potential outcome.

BRIEF DESCRIPTION OF DRAWINGS

[0005] Implementations are described with reference to the following figures, in which:

[0006] FIG. **1** depicts a schematic block diagram of an example computing device configured processing a decision tree in a parallel operation.

[0007] FIG. **2** depicts a schematic block diagram of a row of processing elements of the computing device of FIG. **1**.

[0008] FIGS. **3**A and **3**B depict schematic diagrams of a matrix-vector multiplication and memory loading and processing of the matrix-vector multiplication in an array of processing elements of the computing device of FIG. **1**.

[0009] FIG. **4** depicts a schematic diagram of an example decision tree for processing by the computing device of FIG. **1**.

[0010] FIG. **5** depicts a flowchart of an example method of processing a decision tree in a parallel operation.

[0011] FIG. **6** depicts a flowchart of an example method of initializing a set of processing elements during the method of FIG. **5**.

[0012] FIG. **7** depicts a schematic diagram of an example initialization state of a set of processing elements of the computing device of FIG. **1** during block **505** of the method of FIG. **5**.

[0013] FIGS. **8**A and **8**B depict schematic diagrams of further example states of the set of processing elements of the computing device of FIG. **1** during blocks **510** and **515** of the method of FIG. **5**.

[0014] FIG. **9** depicts a flowchart of an example method of identifying a determined outcome during the method of FIG. **5**.

DETAILED DESCRIPTION

[0015] Decision trees are typically processed sequentially, with certain nodes being processed based on the results of earlier level nodes, resulting in slow and inefficient processing. In particular, in a distributed parallel computing device, the sequential processing can be difficult to manage based on memory and time requirements to store results from earlier level nodes.

[0016] Accordingly, the present exemplary computing device includes a distributed parallel computing system which is configured to process the decision tree on a given input vector with a corresponding distributed parallel processing method. In particular, the computing device may, in a first phase, perform a node comparison operation for the given input vector, wherein each node is processed by one processing element of a bank of processing elements. In particular, each node of the decision tree is processed in parallel, irrespective of the level on which the node resides, or results from earlier level nodes. In a second phase, the result vector from the node comparison operation is accumulated with each outcome vector representing each potential outcome to obtain respective outcome metrics for each potential outcome. In particular, the accumulation of the outcome metrics of the potential outcomes may similarly be performed in parallel, for example using a generalized matrix-vector multiplication in the distributed parallel computing system. The outcome metrics may then be assessed (e.g., including further manipulations and/or normalization) to identify one of the potential outcomes as the determined outcome for the given input vector.

[0017] FIG. **1** shows such an example a computing device **100**. The computing device **100** includes a plurality of banks **102** of processing elements. The banks **102** may be operated in a cooperative manner to implement a parallel processing scheme, such as a SIMD scheme.

[0018] The banks **102** may be arranged in a regular rectangular grid-like pattern, as illustrated. For sake of explanation, relative directions mentioned herein will be

referred to as up, down, vertical, left, right, horizontal, and so on. However, it is understood that such directions are approximations, are not based on any particular reference direction, and are not to be considered limiting. Any practical number of banks **102** may be used. Limitations in semiconductor fabrication techniques may govern. In some examples, 512 banks **102** are arranged in a 32-by-16 grid.

[0019] A bank **102** may include a plurality of rows **104** of processing elements (PEs) **108** and a controller **106**. A bank **102** may include any practical number of PE rows **104**. For example, eight rows **104** may be provided for each controller **106**. In some examples, all banks **102** may be provided with the same or similar arrangement of rows. In other examples, substantially all banks **102** are substantially identical. In still other examples, a bank **102** may be assigned a special purpose in the computing device and may have a different architecture, which may omit PE rows **104** and/or a controller **106**. Any practical number of PEs **108** may be provided to a row **104**. For example, 256 PEs may be provided to each row **104**. Continuing the numerical example above, 256 PEs provided to each of eight rows **104** of 512 banks **102** means the computing device **100** includes about 1.05 million PEs **108**, less any losses due to imperfect semiconductor manufacturing yield. A PE **108** may be configured to operate at any practical bit size, such as one, two, four, or eight bits. PEs may be operated in pairs to accommodate operations requiring wider bit sizes.

[0020] Instructions and/or data may be communicated to/from the banks **102** via an input/output (I/O) bus **110**. The I/O bus **110** may include a plurality of segments. A bank **102** may be connected to the I/O bus **110** by a vertical bus **112**. Additionally or alternatively, a vertical bus **112** may allow communication among banks **102** in a vertical direction. Such communication may be restricted to immediately vertically adjacent banks **102** or may extend to further banks **102**. A bank **102** may be connected to a horizontally neighboring bank **102** by a horizontal bus **114** to allow communication among banks **102** in a horizontal direction. Such communication may be restricted to immediately horizontally adjacent banks **102** or may extend to further banks **102**.

[0021] Communications through any or all of the busses **110**, **112**, **114** may include direct memory access (DMA) to memory of the rows **104** of the PEs **108**. Additionally or alternatively, such communications may include memory access performed through the processing functionality of the PEs **108**.

[0022] The computing device **100** may include a main processor (not shown) to communicate instructions and/or data with the banks **102** via the I/O bus **110**, manage operations of the banks **102**, and/or provide an I/O interface for a user, network, or other device. The I/O bus **110** may include a Peripheral Component Interconnect Express (PCIe) interface or similar.

[0023] FIG. **2** shows an example row **104** including an array of processing elements **108**, which may be physically arranged in a linear pattern (e.g., a physical row). Each PE **108** includes an arithmetic logic unit (ALU) to perform an operation, such as addition, multiplication, and so on. The PEs **108** are mutually connected to share or communicate data. For example, interconnections **200** may be provided among the array of PEs **108** to provide direct communication among neighboring PEs **108**. The interconnections **200** among the PEs **108** and with the controller **106** are shown schematically for sake of explanation, however the interconnections **200** may be direct connections between two given PEs **108** (e.g., between a given PE **108** and a neighboring PE **108**, an n+1 neighbor PE **108**, etc.). The endmost PEs **108** at one end of a row **104** may have connections to a controller **106**. Additionally or alternatively, end-most PEs **108** of one bank **102** may connect in the same relative manner through the controller **106** and to PEs **108** of an adjacent bank **102**. That is, the controller **106** may be connected between two rows **104** of PEs **108** in adjacent banks **102**. In other examples, a set of interconnections **200** may be provided to connect PEs **108** in up-down (column-based) connections, so that information may be shared directly between PEs **108** that are in adjacent rows.

[0024] A row **104** of PEs **108** may include memory **202** to store data for the row **104**. A PE **108** may have a dedicated space in the memory **202**. For example, each PE **108** may be connected to a different range of memory cells **204**. Any practical number of memory cells **204** may be used. In one example, 144 memory cells **204** are provided to each PE **108**.

[0025] The controller **106** may control the array of PEs **108** to perform a SIMD operation with data in the memory **202**. For example, the controller **106** may trigger the PEs **108** to simultaneously add two numbers stored in respective cells **204**.

[0026] The controller **106** may communicate data to and from the memory **202** though the PEs **108**. For example, the controller **106** may load data into the memory **202** by directly loading data into connected PEs **108** and controlling PEs **108** to shift the data to PEs **108** further in the array. PEs **108** may load such data into their respective memory cells **204**. For example, data destined for rightmost PEs **108** may first be loaded into leftmost PEs and then communicated rightwards by interconnections **200** before being stored in rightmost memory cells **204**. Other methods of I/O with the memory, such as direct memory access by the controller **106**, are also contemplated. The memory cells **204** of different PEs **108** may have the same addresses, so that address decoding may be avoided to the extent possible.

[0027] Data stored in memory cells **204** may be any suitable data, such as operands, operators, coefficients, vector components, and similar. In particular, the controller **106** may include a sorting layer **206** configured to select features to be assigned to the row **104** of PEs **108** and to subsequently load suitable operands, operators, coefficients, vector components, and other data to the memory cells **204** according to the selected features and the order in which features are assigned to the PEs **108** as will be further described herein.

[0028] FIG. **3**A shows an example matrix multiplication to be processed by the PEs **108**. A matrix multiplication may be a generalized matrix-vector multiply (GEMV). A matrix multiplication may use a coefficient matrix and an input vector to obtain a resultant vector. In this example, the coefficient matrix is a four-by-four matrix and the vectors are of length four. In other examples, matrices and vectors of any practical size may be used. In other examples, a matrix multiplication may be a generalized matrix-matrix multiply (GEMM).

[0029] FIG. **3**B illustrates an array of PEs **108** and related memory cells **204** for carrying out the matrix multiplication illustrated in FIG. **3**A. Each PE **108** may include local registers **300**, **302** to hold data undergoing an operation. Memory cells **204** may also hold data contributing to the operation.

[0030] As matrix multiplication involves sums of products, the PEs **108** may additively accumulate resultant vector components d_0 to d_3 in respective registers **300**, while input vector components a_0 to a_3 are multiplied by respective coefficients c_{00} to c_{33}. That is, one PE **108** may accumulate a resultant vector component d_0, a neighbor PE **108** may accumulate another resultant vector component d_1, and so on. Resultant vector components d_0 to d_3 may be considered dot products. Generally, a GEMV may be considered a collection of dot products of a vector with a set of vectors represented by the rows of a matrix.

[0031] To facilitate matrix multiplication, the contents of registers **300** and/or registers **302** may be rearranged among the PEs **108**. In this example, resultant vector components d_0 to d_3 remain fixed and input vector components a_0 to a_3 are moved. Further, coefficients coo to c_{33} may be loaded into memory cells to optimize memory accesses.

[0032] In the example illustrated in FIGS. **3**A and **3**B, the input vector components a_0 to a_3 are loaded into a sequence of PEs **108** that are to accumulate resultant vector components d_0 to d_3 in the same sequence. The relevant coefficients c_{00}, c_{11}, c_{22}, c_{33} are accessed and multiplied by the respective input vector components a_0 to a_3. That is, a_0 and c_{00} are multiplied and then accumulated as d_0, a_1 and c_{11} are multiplied and then accumulated as d_1, and so on.

[0033] The input vector components a_0 to a_3 are then rearranged so that a remaining contribution of each input vector components a_0 to a_3 to a respective resultant vector components d_0 to d_3 may be accumulated.

[0034] For example, input vector components a_0 to a_2 are moved one PE **108** to the right and input vector components as is moved three PEs **108** to the left. The result is that a next arrangement of input vector components a_3, a_0, a_1, a_2 at the PEs **108** is achieved, where each input vector component is located at a PE **108** that it has not yet occupied during the present matrix multiplication. Appropriate coefficients c_{03}, c_{10}, c_{21}, c_{32} in memory cells **204** are then accessed and multiplied by the respective input vector components a_3, a_0, a_1, a_2. That is, as and cos are multiplied and then accumulated as d_0, a_0 and c_{10} are multiplied and then accumulated as d_1, and so on.

[0035] The input vector components a_0 to a_3 are then rearranged twice more, with multiplying accumulation being performed with the input vector components and appropriate coefficients at each new arrangement. At the conclusion of four sets of multiplying accumulation and three intervening rearrangements, the accumulated resultant vector components d_0 to d_3 represent the final result of the matrix multiplication.

[0036] Further, the arrangements of coefficients c_{00} to c_{33} in the memory cells **204** may be predetermined, so that each PE **108** may access the next coefficient needed without requiring coefficients to be moved among memory cells **204**. The coefficients c_{00} to c_{33} may be arranged in the memory cells **204** in a diagonalized manner, such that a first row of coefficients is used for a first arrangement of input vector components, a second row of coefficients is used for a second arrangement of input vector components, and so on.

[0037] Hence, the respective memory addresses referenced by the PEs **108** after a rearrangement of input vector components may be incremented or decremented identically. For example, with a first arrangement of input vector components, each PE **108** may reference its respective memory cell at address **0** for the appropriate coefficient. Likewise, with a second arrangement of input vector components, each PE **108** may reference its respective memory cell at address **1** for the appropriate coefficient, and so on.

[0038] Accordingly, the computing device **100** is configured for efficient parallel processing, in particular in performing GEMVs using the banks **102** of PEs **108**. More particularly, in the present example, the computing device **100** may be applied to process a decision tree to identify a determined outcome from an input vector by organizing and converting the decision tree as a matrix of coefficients. The computing device **100** may then perform a GEMV to obtain the determined outcome using parallel instead of sequential processing, as described further herein.

[0039] For example, referring to FIG. **4**, an example decision tree **400** is depicted. The decision tree **400** includes a plurality of nodes **402**; in the present example, five nodes **402***a*-**0**, **402***b*-**1**, **402***b*-**2**, **402***c*-**3**, and **402***c*-**4** are depicted (referred to herein generically as a node **402**, and collectively as nodes **402**; this nomenclature may be used elsewhere herein). In other examples, the decision tree **400** may include more or fewer nodes.

[0040] The decision tree **400** is configured to evaluate an input vector **404** having a plurality of input attributes, i_0, i_1, i_2, i_3, and i_4, where each input attribute corresponds to a feature f. To evaluate the input vector **404**, each node **402** is configured to compare one of the features f to a threshold value t to determine a result at the given node **402**. More particularly, each node **402** is configured to compare a specific input value, or input attribute i for the feature f to the corresponding threshold value t.

[0041] Thus, for example, the node **402***a*-**0** is configured to compare the input attribute i_0 for the feature f_0 to a threshold value t_0, the node **402***b*-**1** is configured to compare the input attribute i_1 for the feature f_1 to a threshold value t_1, the node **402***b*-**2** is configured to compare the input attribute i_2 for the feature f_2 to a threshold value t_2, the node **402***c*-**3** is configured to compare the input attribute is for the feature f_3 to a threshold value t_3, and the node **402***c*-**4** is configured to compare the input attribute i_4 for the feature f_4 to a threshold value t_4.

[0042] The input vector **404** may therefore have at least as many, and preferably the same number, of input attributes **406** as nodes **402** in the decision tree **400**. Thus, in the present example, the input vector **404** has five input attributes; in other examples, the input vector **404** may have more or fewer input attributes.

[0043] As a result of evaluating the input attributes of the input vector **404** at at least a subset of the nodes **402**, a determined outcome o for the input vector **404** may be determined. For example, the decision tree **400** may have six possible outcomes o_0, o_1, o_2, o_3, o_4, and o_5. Evaluation of the input vector **404** by the decision tree **400** allows a single one of the six possible outcomes to be identified based on the combination of results at the nodes **402**.

[0044] The decision tree **400** may be applied in healthcare, business, finance, or the like to objectively evaluate cases (e.g., individuals, organizations, etc.) by comparing certain attributes (i.e., the input attributes) of certain features to fixed thresholds at each of the nodes **402**. For example, the decision tree **400** may evaluate a person's age (e.g., an input attribute of 46 for a specific person, for 'age' as the feature f) to a given age threshold t, a person's income to a given income threshold, etc.

[0045] In a typical application, the decision tree **400** may have a number of levels, of which three example levels **410***a*, **410***b*, and **410***c* are depicted, with each level **410***x* containing corresponding nodes **402***x*. The decision tree **400** may be evaluated sequentially by level **410**, with one node **402** being evaluated in each level **410** of the tree **400**. For example, the node **402***a*-**0** may be evaluated first (i.e., at the first level **410***a*), one of the nodes **402***b*-**1** or **402***b*-**2** may be evaluated next (i.e., at the second level **410***b*) depending on the result at the first level **410***a*, and so on. Accordingly, the nodes **402** are evaluated sequentially and selectively, depending on the result at an earlier level node. When the decision tree **400** includes many nodes **402**, and in particular, many levels of nodes **402**, the evaluation of the decision tree **400** may be slow and consume memory space to track the results at prior nodes as well as input attributes which may or may not be processed at future nodes. In particular, such a sequential and memory-intensive operation may be inefficiently processed on a computing device with distributed, parallel processing, such as the computing device **100**.

[0046] Hence, in accordance with the present disclosure, the decision tree **400** may be parallelized for evaluation by the computing device **100**. FIG. **5** depicts a flowchart of an example method **500** of evaluating a decision tree in a parallel operation to identify a determined outcome for a given input. The method **500** will be described in conjunction with its performance by the computing device **100**, with reference to the decision tree **400**. In other examples, other suitable devices and/or decision trees may be employed. In some examples, some or all of the blocks of the method **500** may be performed in an order other than that depicted, including simultaneous performance of some of the blocks, or similar.

[0047] At block **505**, the computing device **100** is configured to initialize the bank(s) **102** and/or rows **104** of PEs **108** and the associated memory cells **204** for evaluating an input vector (such as the input vector **404**) according to the decision tree **400**. In particular, the computing device **100** may load data into the memory cell **204** corresponding to each PE **108** for subsequent node comparison and outcome evaluation operations, as will be described in further detail below. In some examples, the computing device **100** may further load data into the memory cell **204** corresponding to each PE **108** for an outcome selection operation, as described in further detail below. In some examples, a subset of the features or nodes of the decision tree **400** may be selected by the sorting layer **206** for processing by a given bank **102** and/or row **104** of PEs **108**.

[0048] For example, referring to FIG. **6**, a flowchart of an example method **600** of initializing the decision tree **400** the computing device **100** is depicted. In particular, the method **600** systematizes the decision tree **400** and structures the computing device **100** to allow the computing device **100** to efficiently evaluate the decision tree **400**.

[0049] At block **605**, the computing device **100** is configured to order the nodes **402** of the decision tree **400**. Subsequent operations to evaluate the decision tree **400** may use the nodes to assign various coefficients, thresholds, and/or other data to each of the nodes **402**, and accordingly, the computing device **100** may order the nodes **402** to better track the nodes **402**. In some examples, the computing device **100** may assign an identifier to each node **402**. For example, with reference to the decision tree **400**, the computing device **100** may order the nodes **402** as follows: [**402***a*-**0**, **402***b*-**1**, **402***b*-**2**, **402***c*-**3**, **402***c*-**4**].

[0050] Ordering the nodes **402** may allow the computing device **100** to treat the nodes **402** substantially equally and in parallel, rather than hierarchically (and therefore sequentially) based on the result from a node in an earlier level **410**, as will be described further herein. Thus, in another example, the nodes **402** may be ordered as follows: [**402***c*-**3**, **402***a*-**0**, **402***c*-**4**, **402***b*-**2**, **402***b*-**1**], or any other order (e.g., randomly or systematically ordered) which is fixed after the performance of block **605**.

[0051] At block **610**, the computing device **100** assigns and/or maps each node **402** to one PE **108** for the node comparison operation. That is, the assigned and/or mapped PE **108** represents the node **402** and is configured to perform the comparison of an input attribute to a corresponding threshold value. Thus, the comparison at each node **402** is performed at a separate PE **108** in one or more of the banks **102** and may be computed in parallel.

[0052] Accordingly, at block **615**, the computing device **100** is configured to load the threshold t for each node **402** into the associated memory cell **204** of the PE **108** configured to evaluate the node **402**. In particular, the threshold values t may be stored at the same address in each of the memory cell **204** to allow the node comparison operation to be parallelized across the PEs **108**.

[0053] For example, referring to FIG. **7**, a partial representation of an array of PEs **108** is depicted. In particular, FIG. **7** depicts an initialization state **700** at initialization of the decision tree **400** to be evaluated by the computing device **100** (e.g., after performance of the method **600**). In particular, six PEs **108** are depicted, namely, PEs **108**-**0**, **108**-**1**, **108**-**2**, **108**-**3**, **108**-**4**, and **108**-**5**.

[0054] Accordingly, at block **610**, the computing device **100** assigns each node **402** to one of the PEs **108**. In the present example, the node **402***a*-**0** is assigned to the PE **108**-**0**, the node **402***b*-**1** is assigned to PE **108**-**1**, the node **402***b*-**2** is assigned to the PE **108**-**2**, the node **402***c*-**3** is assigned to the PE **108**-**3**, the node **402***c*-**4** is assigned to the PE **108**-**4**. Since there are fewer nodes **402** than PEs **108**, the PE **108**-**5** may remain unassigned for the node comparison operation.

[0055] Subsequently, at block **615**, the computing device **100** may load the thresholds t into the corresponding memory cells **204** of the assigned PEs **108**. In the present example, the threshold to for the node **402***a*-**0** is loaded into a memory cell **204**-**0** corresponding to PE **108**-**0**, the threshold t_1 for the node **402***b*-**1** is loaded into a memory cell **204**-**1** corresponding to PE **108**-**1**, the threshold t_2 for the node **402***b*-**2** is loaded into a memory cell **204**-**2** corresponding to PE **108**-**2**, the threshold t_3 for the node **402***c*-**3** is loaded into a memory cell **204**-**3** corresponding to PE **108**-**3**, and the threshold t_4 for the node **402***c*-**4** is loaded into a memory cell **204**-**4** corresponding to PE **108**-**4**.

[0056] Returning to FIG. **6**, at block **620**, the computing device **100** is configured to obtain outcome vectors for each possible outcome o of the decision tree **400**. In particular, each possible outcome o may be represented by a corresponding outcome vector, the components of which represent the result at each node **402** in order to obtain the possible outcome o. The components of the outcome vector may be ordered according to the order of the nodes defined

at block **605**, such that a given coefficient of each outcome vector represents the different results for each outcome o at the same node **402**.

[0057] For example, for the first example order of nodes **402** given above, the outcome vectors may be represented by: [c_{00}, c_{o1}, c_{o2}, c_{o3}, c_{o4}], where each Con represents the result at the corresponding node n to obtain a given outcome o. The results (or coefficients) may be represented by a value of 1 if the result at the given node **402** is "Yes" or affirmative (i.e., that the input attribute exceeds the threshold value for the given node **402**), a value of −1 if the result at the given node **402** is "No" or negative (i.e., that the input attribute does not exceed the threshold value for the given node **402**), and a value of 0 if the result at the given node **402** is immaterial or not evaluated based on the result of another node **402** at an earlier level.

[0058] For example, the outcome vector for the outcome o_1 may be given as follows [1, 1, 0, −1, 0]. That is, in order to arrive at the outcome o_1, the result at the node **402***a*-**0** is affirmative (1), the result at the node **402***b*-**1** is affirmative (1), the result at the node **402***b*-**2** is immaterial given the affirmative result at the (earlier) node **402***a*-**0** (0), the result at the node **402***c*-**3** is negative (−1), and the result at the node **402***c*-**4** is immaterial given the affirmative result at the (earlier) node **402***b*-**1** (0).

[0059] At block **625**, the computing device **100** assigns and/or maps each possible outcome o to one PE **108** for the outcome evaluation operation. That is, the assigned and/or mapped PE **108** represents the possible outcome o and is configured to accumulate an outcome metric representing the possible outcome o. In particular, the outcome metric is accumulated based on the outcome vector for the possible outcome o and a result vector from the node comparison operation. For example, the outcome metric may be a dot product of the outcome vector with the result vector. Thus, the evaluation of each of the possible outcomes o is performed at a separate PE **108** in one or more of the banks **102** and/or rows **104** and may be computed in parallel.

[0060] Accordingly, at block **630**, the computing device **100** is configured to load the coefficients for the outcome vectors into the associated memory cells **204** of the PEs **108**. In some examples, each outcome vector may be loaded into the corresponding memory cell **204** for the PE **108** configured to evaluate the outcome vector. In other examples, the coefficients of the outcome vectors may be loaded into the memory cells **204** in a diagonalized and/or otherwise patterned manner to allow the performance of a GEMV as described above by rotating and/or rearranging the input vector components (i.e., the components of the result vector) and accumulating the results in the local register. That is, the computing device **100** may define an outcome matrix defined based on the outcome vectors. The coefficients of the outcome matrix may then be loaded into the memory cells **204** in such a manner that each PE **108** accumulates the dot product for its corresponding assigned outcome vector with the result vector.

[0061] Referring again to FIG. **7**, the computing device **100** may assign, at block **625**, each outcome o to one of the PEs **108**. In the present example, the outcome o_0 is assigned to the PE **108-0**, the outcome o_1 is assigned to the PE **108-1**, the outcome o_2 is assigned to the PE **108-2**, the outcome o_3 is assigned to the PE **108-3**, the outcome o_4 is assigned to the PE **108-4**, and the outcome o_5 is assigned to the PE **108-5**.

[0062] Subsequently, at block **630**, the computing device **100** may load the outcome vectors representing the outcomes o into the memory cells **204**. In the present example, for simplicity, the outcome vector having coefficients c_{00} through c_{o4} representing the outcome o_0 is loaded into the memory cell **204-0**, the outcome vector having components c_{10} through c_{14} representing the outcome o_1 is loaded into the memory cell **204-1**, the outcome vector having components c_{20} through c_{24} representing the outcome o_2 is loaded into the memory cell **204-2**, the outcome vector having components c_{30} through c_{34} representing the outcome o_3 is loaded into the memory cell **204-3**, the outcome vector having components c_{40} through c_{44} representing the outcome o_4 is loaded into the memory cell **204-4**, and the outcome vector having components c_{50} through c_{54} representing the outcome o_5 is loaded into the memory cell **204-5**.

[0063] As noted above, in other examples, the coefficients of the outcome vectors for the outcomes o may be extracted from the outcome vectors and loaded into the memory cells **204** in a diagonalized (or otherwise suitably patterned) manner based on the predefined rearrangements of the components in the local registers of the PEs **108**. In such examples, the coefficients of the outcome vectors are loaded into the memory cells **204** such that the outcome metric for the outcome o_0 is accumulated in the PE **108-0**, the outcome metric for the outcome o_1 is accumulated in the PE **108-1**, the outcome metric for the outcome o_2 is accumulated in the PE **108-2**, the outcome metric for the outcome o_3 is accumulated in the PE **108-3**, the outcome metric for the outcome o_4 is accumulated in the PE **108-4**, and the outcome metric for the outcome o_5 is accumulated in the PE **108-5**.

[0064] Returning again to FIG. **6**, at block **635**, the computing device **100** may be configured to load a depth value for each outcome into the associated memory cell **204** of the PE **108** assigned to accumulate the outcome metric for the outcome. In particular, each outcome is associated with a certain depth or number of levels, or number of nodes to be evaluated to reach the given outcome. For example, the outcomes o_0, o_1, o_2, and o_3 have depths of three—that is, there are three levels **410***a*, **410***b*, and **410***c* for which a node comparison is performed to reach the outcome. The outcomes o_4 and o_5 have depths of two—that is, there are two levels **410***a* and **410***b* for which a node comparison is performed to reach the outcome.

[0065] The depth of each outcome therefore represents the number of nodes **402** which are evaluated to reach the outcome. Accordingly, during the outcome selection operation, the number of matching node comparison results may differ for different nodes, and therefore the depth of each outcome affects the outcome selection operation. Since the depths may be different for different outcomes, the computing device **100** loads a depth value for each outcome into the associated memory cell **204** of the PE configured to accumulate the outcome metric for the given outcome.

[0066] In some examples, the depth value may directly correspond to the depth for the outcome. In other examples, to simplify the outcome selection operation, the depth value may correspond to the depth minus one, or another suitable predefined formulaic relationship to the depth.

[0067] At block **640**, the computing device **100** may be configured to load an outcome identifier for each outcome into the associated memory cell **204** of the PE **108** assigned to accumulate the outcome metric for the outcome. In particular, the outcome identifier may preferably be a

numeric identifier to uniquely identify each of the outcomes during the outcome selection operation, as will be described in further detail herein.

[0068] For example, referring again to FIG. **7**, the computing device **100** may load the depth value dv_0 and the outcome identifier z_0 into the memory cell **204-0**, the depth value dv_1 and the outcome identifier z_1 into the memory cell **204-1**, the depth value dv_2 and the outcome identifier z_2 into the memory cell **204-2**, the depth value dv_3 and the outcome identifier z_3 into the memory cell **204-3**, the depth value dv_4 and the outcome identifier z_4 into the memory cell **204-4**, the depth value dv_3 and the outcome identifier z_5 into the memory cell **204-5**.

[0069] Returning now to FIG. **5**, after completing initialization of the PEs **108** to evaluate the decision tree **400**, at block **510**, the computing device **100** obtains an input vector, such as the input vector **404**, to be processed by the decision tree **400**. The input vector **404** includes the input attributes (i.e., the specific values for a given person, organization, case, etc. for the respective features) to be evaluated at each of the nodes **402**. The input vector **404** may be loaded into the PEs **108** (or a subset of the PEs **108**), for example in the local register **302** of the PEs **108**. In particular, the computing device **100** may determine, for each given input attribute i, the corresponding node **402** configured to evaluate the given input attribute and load the given input attribute i into the local register **302** of the PE **108** assigned to the corresponding node **402**. Thus, each input attribute of the input vector **404** may be loaded into a separate PE **108**.

[0070] For example, the sorting layer **206** may be configured to sort the input attributes of the input vector **404** according to the nodes **402** assigned to a given set of PEs **108** (e.g., in a bank **102** or a row **104**), and in particular, each input attribute to the particular node **402** configured to evaluate the input attribute. Preferably, the input vector **404** may be structured according to the order of nodes **402** identified at block **605** of the method **600** to facilitate loading the input vector **404** to the PEs **108**. In other examples, other manners of sorting the input attributes of the input vector **404** are also contemplated, for example by using a "one-hot" matrix of coefficients, in which each PE **108** in the sorting layer **206** would include a single "1" coefficient for a given target feature to sort through a set of features to create the input vector **404**.

[0071] At block **515**, the computing device **100** is configured to perform the node comparison operation, in which the PEs **108** process the input vector to obtain a result vector. In particular, each PE **108** may compare the respective input attribute stored in the local register **302** to the threshold value stored at the predefined address in the associated memory cell **204**. Since the input attributes and threshold values are loaded into the local registers **302** and memory cells **204** based on the node **402** assigned to a given PE **108**, each PE **108** is configured to perform the comparison at the corresponding node **402**. Further, the computing device **100** controls the PEs **108** to evaluate (i.e., to perform the assigned comparison of the input attribute to the threshold) each of the nodes **402** in the decision tree **400** in parallel, irrespective of the level **410** of the node **402** or the results of earlier nodes **402** in earlier levels **410**.

[0072] Thus, the node comparison operation at block **515** represents a first phase of processing the input vector **404**. As a result of the node comparison operation at block **515**, the computing device **100** obtains a result vector representing the results at each of the nodes **402** of the decision tree **400**. That is, the result vector is composed of the respective result of the node comparison operation obtained at each PE **108**.

[0073] Additionally at block **515**, the computing device **100** may store the result vector. In some examples, the computing device **100** may store result vector of the node comparison operation by replacing each of the input attributes for a given node **402** with the result for that node (i.e., the result of the comparison of the input attribute i with the corresponding threshold t) in the local register **302**. For example, an affirmative result may be stored as a value of 1, while a negative result may be stored as a value of −1.

[0074] For example, referring to FIG. **8**A, a partial representation of the array of PEs **108** in states **800** after obtaining the input vector **404** is depicted. In particular, after obtaining the input vector **404**, the local registers **302** are loaded with the input attributes i corresponding with the node-PE assignment. FIG. **8**B depicts the array of PEs **108** in a state **810** after completion of the node comparison operation, for example as described at block **515**. Specifically, the computing device **100** completes the node comparison operation to compare the value of the input attribute i stored in the local register **302** with the corresponding threshold t stored at the predetermined memory address in the memory cell **204**. After completing the node comparison operation, each PE **108** may replace the input attribute i in the local register **302** with the result r of the comparison. In particular, the PE **108** may update the local register **302** with a value of 1 if the input attribute i exceeds the threshold t, and a value of −1 if the input attribute i does not exceed the threshold t.

[0075] Returning again to FIG. **5**, at block **520**, the computing device **100** is configured to perform the outcome evaluation operation. In particular, each PE **108** is configured to accumulate the result vector from the node comparison operation with a respective outcome vector to obtain a respective outcome metric for the outcome assigned to the PE **108**. The outcome metric may be a dot product of the result vector and the respective outcome vector. In particular, the coefficients of the outcome vectors are loaded into the memory cells **204** to allow the computing device **100** to perform a GEMV including multiple iterations of multiplications, accumulations and rotations to allow each PE **108** to accumulate the dot product of the result vector with the outcome vector for the outcome assigned to the PE **108** (i.e., the outcome metric for the outcome assigned to the PE **108**). Accordingly, the accumulations of the respective outcome metrics at each of the PEs **108** may be performed in parallel, and the outcome metrics for each of the outcomes may be subsequently assessed in parallel, irrespective of the specific results at any individual node **402**.

[0076] Each PE **108** may be configured to store the accumulations, including the intermediate accumulations during performance of the GEMV, in the local register **300**. Accordingly, upon completion of the GEMV and the outcome evaluation operation, the local registers **300** of the PEs **108** may store the resultant outcome metrics for the respective outcomes assigned to the PEs **108**.

[0077] At block **525**, the computing device **100** is configured to identify or select one of the potential outcomes as the determined outcome of the decision tree for the input vector based on the respective outcome metrics of the potential outcomes. That is, the computing device **100** is configured to analyze the outcome metrics obtained at block **520** from

the outcome evaluation operation to select one of the potential outcomes as the determined outcome. For example, the determined outcome may be identified as the potential outcome having the largest outcome metric, the outcome metric having a specific predetermined value, or the computing device **100** may apply further manipulations or other suitable assessments of the outcome metrics, based for example on the manner in which the coefficients of the outcome vectors and the result vectors are assigned.

[0078] For example, FIG. **9** depicts an example method **900** of assessing the outcome metrics to select one of the potential outcomes as the determined outcome.

[0079] At block **905**, the computing device **100** may be configured to compare the outcome metric for each outcome to the corresponding depth value stored in the associated memory cell **204** of the PE **108** assigned to accumulate the outcome metric.

[0080] In particular, in the present example, the outcome metrics are computed as dot products between an outcome vector for a given outcome and the result vector. Both the outcome vector and the result vector are represented by coefficients with a value of 1 for an affirmative result at a given node, and a value of −1 for a negative result at a given node. Additionally, the outcome vector includes coefficients with a value of 0 for nodes whose results are irrelevant for the given outcome.

[0081] Accordingly, when the results at a given node match (i.e., are either both 1 or are both−1), then the product of the outcome coefficient and the result coefficient is positive and provides a positive contribution to the outcome metric. When the results at a given node do not match (i.e., one has a value of 1 and the other a value of −1, or the node is irrelevant with a value of 0), then the product of the outcome coefficient and the result coefficient is zero or negative and provides no contribution or a negative contribution to the outcome metric.

[0082] Specifically, in order to reach a given outcome (i.e., the true outcome), the result vector will match the outcome vector at each level **410** in the decision tree, and accordingly, the outcome should have a resulting outcome metric equal to the depth associated with the outcome. Accordingly, in some examples, at block **905**, the computing device **100** may compare the outcome metric directly to the depth stored as the depth value.

[0083] In other examples, to perform the comparison of the outcome metric to the depth value, the computing device **100** may subtract the stored depth value from the outcome metric. In the case where the depth value is equal to the depth for the respective outcome, the result of the subtraction is zero for the true outcome, and a negative value for the other outcomes. In the case where the depth value is equal to the depth less one for the respective outcome, the result of the subtraction is one for the true outcome and zero or a negative value for the other outcomes.

[0084] At block **910**, the computing device **100** is configured to normalize the outcome metric. For example, the computing device **100** may store the result of the comparison at block **905**, for example as an updated outcome metric. In particular, the computing device **100** may be configured to store the result of the comparison in the local register **300**, replacing the previously computed outcome metric. For example, the computing device **100** may store the result of the subtraction performed as the comparison, or may conditionally store a value of one if the outcome metric is equal to the depth, and a value of zero if the outcome metric is not equal to the depth.

[0085] In some examples, at block **910**, the computing device **100** may additionally normalize the result of the comparison. For example, where the comparison is a subtraction, the computing device **100** may perform a rectified linear unit (RELU) to set negative results to zero. This may be particularly useful when the depth value is assigned to be the depth less one for the respective outcome, such that the result of the subtraction is one for the true outcome and is normalized to zero for the other outcomes.

[0086] At block **915**, the computing device **100** is configured to multiply the updated outcome by the outcome identifier for each outcome stored in the memory cell **204** associated with the PE **108** for the outcome. The result of the multiplication may similarly be stored as a further updated outcome metric. In particular, the true outcome is preferably the only outcome having an updated outcome metric of one, while the other outcomes are preferably set to zero. Thus, after the performance of block **915**, the true outcome may remain the sole outcome having a non-zero updated outcome metric stored in the local register **300** of the PE **108** assigned to the outcome. Specifically, the updated outcome metric may be equal to the value of the outcome identifier for the true outcome.

[0087] At block **920**, the computing device **100** returns the outcome identifier to identify the determined outcome of the decision tree **400** for the input vector **404**. For example, the computing device **100** may determine the largest or maximum value stored in the local registers **300** of the PEs **108**, and return that value as the outcome identifier for the determined outcome. In other examples, the computing device **100** may sum the values stored in the local registers **300** of the PEs **108** and may return the sum as the outcome identifier for the determined outcome.

[0088] Accordingly, as described herein, a computing device and method are provided to allow processing of a decision tree on an input vector with parallel processing. In a first phase, each of the attributes of the input vector is evaluated, representing the evaluation at each of the nodes of the decision tree. In particular, in the first phase, each node and attribute is processed, irrespective of the level of the node or results from earlier levels, thereby bypassing the natural hierarchy of the decision tree. In a second phase, the result vector is accumulated with a plurality of outcome vectors, each outcome vector representing one potential outcome of the decision tree. The potential outcomes are then represented by the outcome metrics (i.e., the results of the accumulations). Similarly, in the second phase, each potential outcome is compared with the result vector, irrespective of results from any of the individual nodes, similarly bypassing the natural hierarchy of the decision tree. The outcome metrics may then be assessed or analyzed to

identify one of the potential outcomes as the determined or true outcome for the application of the decision tree on the input vector.

[0089] The scope of the claims should not be limited by the embodiments set forth in the above examples but should be given the broadest interpretation consistent with the description as a whole.

1. A computing device comprising:

a bank of processing elements;

a controller interconnected with the bank of processing elements, the controller configured to:

obtain an input vector having a plurality of input attributes, the input vector to be processed by a decision tree to identify a determined outcome for the input vector;

control the bank of processing elements to process the input vector to obtain a result vector, wherein each input attribute is processed by one of the processing elements in the bank to obtain a result, and wherein the result vector comprises a combination of the results;

control the bank of processing elements to accumulate the result vector with an outcome vector for each potential outcome of the decision tree to obtain a respective outcome metric for each potential outcome; and

select one potential outcome as the determined outcome of the decision tree for the input vector based on the respective outcome metrics for each potential outcome.

2. The computing device of claim **1**, wherein the decision tree comprises a plurality of nodes, each node configured to process a given input attribute of the input vector.

3. The computing device of claim **2**, wherein the controller is further configured to: assign each node of the decision tree to one of the processing elements to process the given input attribute to obtain the result.

4. The computing device of claim **1**, wherein to process the input attribute, the processing element is configured to compare the input attribute to a predefined threshold for the input attribute.

5. The computing device of claim **4**, wherein the controller is further configured to initialize the bank of processing elements to store the predefined threshold for each input attribute in a respective corresponding memory cell of the processing element.

6. The computing device of claim **1**, wherein the controller is configured to assign each respective outcome metric to be accumulated by one of the processing elements in the bank.

7. The computing device of claim **1**, wherein the respective outcome metric comprises a dot product between the result vector and the respective outcome vector.

8. The computing device of claim **7**, wherein the controller is configured to apply a generalized matrix-vector multiply between the result vector and an outcome matrix comprising the outcome vectors to accumulate the respective outcome metrics.

9. The computing device of claim **8**, wherein the controller is configured to initialize the bank of processing elements to load the outcome matrix into memory cells associated with the processing elements.

10. The computing device of claim **1**, wherein to select the determined outcome, the controller is configured to:

compare each outcome metric to a depth value for the potential outcome; and

normalize the outcome metrics;

multiply each normalized outcome metric by a respective outcome identifier; and

return the outcome identifier identifying the determined outcome.

11. A method comprising:

obtaining an input vector having a plurality of input attributes, the input vector to be processed by a decision tree to identify a determined outcome for the input vector;

controlling a bank of processing elements to process the input vector to obtain a result vector, wherein each input attribute is processed by one of the processing elements in the bank to obtain a result, and wherein the result vector comprises a combination of the results;

controlling the bank of processing elements to accumulate the result vector with an outcome vector for each potential outcome of the decision tree to obtain a respective outcome metric for each potential outcome; and

selecting one potential outcome as the determined outcome of the decision tree for the input vector based on the respective outcome metrics for each potential outcome.

12. The method of claim **11**, wherein the decision tree comprises a plurality of nodes, each node configured to process a given input attribute of the input vector.

13. The method of claim **12**, further comprising: assigning each node of the decision tree to one of the processing elements to process the given input attribute to obtain the result.

14. The method of claim **11**, wherein processing the input attribute comprises comparing the input attribute to a predefined threshold for the input attribute.

15. The method of claim **14**, further comprising initializing the bank of processing elements to store the predefined threshold for each input attribute in a respective corresponding memory cell of the processing element.

16. The method of claim **11**, further comprising assigning each respective outcome metric to be accumulated by one of the processing elements in the bank.

17. The method of claim **11**, wherein the respective outcome metric comprises a dot product between the result vector and the respective outcome vector.

18. The method of claim **17**, further comprising applying a generalized matrix-vector multiply between the result vector and an outcome matrix comprising the outcome vectors to accumulate the respective outcome metrics.

19. The method of claim **18**, further comprising initializing the bank of processing elements to load the outcome matrix into memory cells associated with the processing elements.

20. The method of claim **11**, wherein selecting the determined outcome comprises:

comparing each outcome metric to a depth value for the potential outcome; and

normalizing the outcome metrics;

multiplying each normalized outcome metric by a respective outcome identifier; and

returning the outcome identifier identifying the determined outcome.

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